

GCP DAY 2: STORAGE AND DATABASE SERVICES

Comprehensive Tutorial and Lab Guide

Learning Objectives

Upon completing this training, participants will be able to:

- Understand Google Cloud Storage fundamentals and console operations
- Master Private Google Access and Cloud NAT configurations
- Learn Cloud SQL for MySQL deployment and management
- Explore Cloud Spanner for distributed SQL databases
- Work with BigTable for NoSQL use cases
- Master gsutil command-line operations for Cloud Storage

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1. INTRODUCTION TO GCP STORAGE AND DATABASE SERVICES

Google Cloud Platform provides a comprehensive suite of storage and database services designed to meet diverse application requirements. Day 2 of your GCP learning journey focuses on the core storage and database offerings.

Service Overview

Cloud Storage (GCS)

Object storage service for unstructured data with high availability and scalability. Best for files, backups, media, and archives at any scale.

Cloud SQL

Fully managed relational database service supporting MySQL, PostgreSQL, and SQL Server. Perfect for OLTP and transactional workloads requiring automated operations.

Cloud Spanner

Globally distributed relational database with strong consistency. Ideal for global applications requiring ACID transactions and massive throughput.

BigTable

NoSQL wide-column store for massive-scale analytical and operational workloads. Optimized for time-series data and sparse data structures.

Networking Services

Private Google Access and Cloud NAT enable secure, private connectivity between your resources and Google services and the internet.

Service Comparison Matrix

Service	Type	Use Case	Consistency	Scale
GCS	Object Storage	Files, backups	Strong	Petabyte+
Cloud SQL	Relational DB	OLTP transactions	Strong	Terabyte
Spanner	Global SQL DB	Global transactions	Strong	Petabyte+
BigTable	NoSQL DB	Analytics, time-series	Eventual	Petabyte+

2. LAB 1: CONFIGURING PRIVATE GOOGLE ACCESS AND CLOUD NAT

Overview

Private Google Access and Cloud NAT are critical components for securing your GCP infrastructure. They allow your internal resources to communicate with Google services and the internet without exposing their internal IP addresses.

Private Google Access

Private Google Access allows VMs on private networks to reach Google APIs without internet access. It uses internal IP addresses to connect to Google services, improving security by eliminating the need for external IPs for API calls. This requires DNS configuration for googleapis.com.

Cloud NAT

Cloud NAT (Network Address Translation) provides network address translation for outbound traffic. It allows private VMs to initiate connections to external resources while not accepting inbound traffic. Cloud NAT provides static public IPs for outbound connections.

Lab Instructions: Step-by-Step Guide

PART A: Enable Private Google Access

Step 1:

Navigate to VPC Networks > Subnets in Google Cloud Console

Step 2:

Select the subnet where you want to enable Private Google Access

Step 3:

Click the Edit button in the subnet details

Step 4:

Locate and toggle the Private Google Access switch to ON

Step 5:

Click Save and wait 1-2 minutes for configuration to complete

PART B: Configure Cloud NAT

Step 1:

Navigate to Networking > Cloud NAT in the console

Step 2:

Click Create Cloud NAT button

Step 3:

Provide Gateway Name: nat-gateway-1

Step 4:

Select your VPC Network from the dropdown

Step 5:

Select the Region where your subnet is located

Step 6:

For Cloud NAT Mapping, select 'All subnets in the region'

Step 7:

For NAT IPs, choose 'Automatic' to let GCP allocate public IPs

Step 8:

Click Create to provision the NAT gateway

Verification Steps

1. Create a test VM without public IP in the configured subnet
2. Connect via SSH using Identity-Aware Proxy
3. Test: curl https://www.googleapis.com (should work with Private Google Access)
4. Test: curl https://www.google.com (should work via Cloud NAT)
5. Monitor Cloud NAT statistics in the console

3. LAB 2: CLOUD STORAGE - QWIK START

Overview

Google Cloud Storage (GCS) is a serverless, secure, and highly scalable object storage service. It's designed to store unstructured data such as images, videos, documents, and archives at any scale.

Key Characteristics

- Bucket-Based Storage: Data organized into buckets at the project level
- Global Namespace: Bucket names must be globally unique across all GCP
- Storage Classes: Standard, Nearline, Coldline, and Archive with different pricing

- Multiple Regions: Choose single, multi-region, or dual-region storage
- Access Control: IAM and ACLs for fine-grained permission management
- Versioning: Enable object versioning for data protection and recovery

Storage Classes Explained

Class	Availability	Min Storage	Cost	Best For
Standard	99.99%	None	High	Active data
Nearline	99.95%	30 days	Lower	Monthly access
Coldline	99.95%	90 days	Even Lower	Quarterly access
Archive	99.95%	365 days	Lowest	Annual access

Lab Instructions: Create and Manage Buckets

PART A: Create a Cloud Storage Bucket

Step 1:

Open Google Cloud Console and navigate to Cloud Storage > Buckets

Step 2:

Click 'Create Bucket' button

Step 3:

Enter globally unique bucket name (lowercase, numbers, hyphens only)

Step 4:

Choose location type: Multi-region (high availability) or Region (single location)

Step 5:

Select 'Standard' storage class for this lab

Step 6:

Choose 'Uniform' access control for simplified management

Step 7:

Click 'Create' to provision your bucket

PART B: Upload and Manage Objects

Step 1:

Click on your newly created bucket from the list

Step 2:

Click 'Upload Files' button

Step 3:

Select a file from your computer or drag and drop

Step 4:

Confirm the upload completes successfully

Step 5:

Click on an uploaded object to view its metadata and properties

PART C: Configure Bucket Settings

Versioning:

Navigate to Bucket Settings and enable versioning for data protection

Lifecycle Policies:

Set policies to automatically delete or archive objects after specified age

Labels:

Add metadata tags for organization and billing purposes

Access Logging:

Enable logging to track all bucket access for compliance and auditing

4. LAB 3: CLOUD SQL FOR MYSQL - QWIK START

Overview

Cloud SQL is a fully managed relational database service that handles patching, backups, replication, and failover automatically. It supports MySQL, PostgreSQL, and SQL Server.

Advantages of Cloud SQL

- Automated backups and point-in-time recovery capabilities
- High availability with automatic failover to replicas
- Automatic software patching and security updates
- Built-in encryption at rest and in transit
- Easy scaling of storage and compute resources
- Integration with Cloud IAM for access control
- Read replicas for scaling read operations

Lab Instructions: Create MySQL Instance

PART A: Create Cloud SQL Instance

Step 1:

Navigate to SQL in the Google Cloud Console

Step 2:

Click 'Create Instance'

Step 3:

Select 'MySQL' as the database engine

Step 4:

Choose 'Development' preset configuration for this lab

Step 5:

Provide Instance ID: mysql-instance-1

Step 6:

Set a secure root password

Step 7:

Select region (same as other resources for latency optimization)

Step 8:

Machine type: db-f1-micro, Storage: SSD 10GB

Step 9:

Click 'Create Instance' and wait 3-5 minutes for provisioning

PART B: Connect and Create Database

Option 1 - Cloud Console (Recommended):

Navigate to your instance and click 'Connect using Cloud Shell'

Create Database:

```
mysql> CREATE DATABASE testdb;
```

```
mysql> USE testdb;
```

Create Table:

```
mysql> CREATE TABLE users (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(255), email VARCHAR(255));
```

Insert Data:

```
mysql> INSERT INTO users (name, email) VALUES ('John Doe', 'john@example.com');
```

Query Data:

```
mysql> SELECT * FROM users;
```

5. LAB 4: CLOUD SPANNER - QWIK START

Overview

Cloud Spanner is a globally distributed, horizontally scalable relational database with ACID transactions and strong consistency. It's designed for mission-critical applications requiring global scale with transactional consistency.

When to Use Cloud Spanner

- Global applications requiring strong consistency
- High transaction throughput at massive scale
- Mission-critical applications where data integrity is essential
- Multi-region deployments with centralized consistency

Lab Instructions: Create Spanner Instance

PART A: Create Instance

Step 1:

Navigate to Spanner in Google Cloud Console

Step 2:

Click 'Create Instance'

Step 3:

Instance Name: spanner-instance-1

Step 4:

Configuration: Choose 'Regional' for single region deployment

Step 5:

Processing Units: Start with 100 PU for the lab

Step 6:

Click 'Create' and wait for provisioning

PART B: Create Database with Schema

Step 1:

Click your instance then 'Create Database'

Step 2:

Database ID: testdb

Step 3:

Define SQL schema in DDL editor:

```
CREATE TABLE Singers ( SingerId INT64 NOT NULL, FirstName STRING(1024), LastName  
STRING(1024), BirthDate DATE, ) PRIMARY KEY(SingerId);
```

Step 4:

Click 'Create' to apply the schema and create the database

PART C: Query Data

Insert Sample Data:

```
INSERT INTO Singers (SingerId, FirstName, LastName) VALUES (1, 'Jane', 'Smith');
```

Query Data:

```
SELECT * FROM Singers;
```

6. LAB 5: BIGTABLE - QWIK START

Overview

BigTable is a massively scalable NoSQL database. It's a distributed, wide-column store optimized for large analytical and operational workloads with billions of rows and millions of columns.

When to Use BigTable

- Time-series data (IoT sensors, stock prices, metrics)
- Massive analytical workloads
- Hierarchical and sparse data structures
- Applications requiring millisecond latency at petabyte scale

Lab Instructions: Create BigTable Instance

PART A: Create Instance

Step 1:

Navigate to BigTable in Google Cloud Console

Step 2:

Click 'Create Instance'

Step 3:

Instance Type: Choose 'Development' for this lab

Step 4:

Instance ID: bigtable-instance-1

Step 5:

Region: Select your preferred region

Step 6:

Nodes: 1 (sufficient for development environment)

Step 7:

Click 'Create' and wait for provisioning

PART B: Create Table and Column Family

Step 1:

Click on your instance from the list

Step 2:

Click 'Create Table'

Step 3:

Table ID: test-table

Step 4:

Create Column Family: cf1 (column family 1)

Step 5:

Click 'Create Table' to complete setup

PART C: Write Data Using Cloud Shell

Step 1:

Open Cloud Shell from the console

Step 2:

Install Python client library:

```
pip install google-cloud-bigtable
```

Step 3:

Create Python script to write data to BigTable

Step 4:

Execute script to insert rows and verify in console

7. LAB 6: USING GSUTIL FOR BUCKET AND OBJECT OPERATIONS

Overview

gsutil is a Python-based command-line tool for interacting with Google Cloud Storage. It enables powerful management of buckets and objects from your terminal at scale.

Installation and Setup

Cloud Shell: Pre-installed and ready to use

Local Installation:

```
curl https://sdk.cloud.google.com | bash
```

Authentication:

```
gcloud auth login
```

Common gsutil Commands

Bucket Management

List All Buckets

```
gsutil ls
```

Create Bucket

```
gsutil mb -l us-central1 gs://my-unique-bucket-name
```

Get Bucket Info

```
gsutil bucketinfo gs://bucket-name
```

Object Operations

Upload Single File

```
gsutil cp local-file.txt gs://bucket-name
```

Upload Directory Recursively

```
gsutil -m cp -r local-directory gs://bucket-name
```

Download File

```
gsutil cp gs://bucket-name/remote-file ./local-path
```

List Objects

```
gsutil ls -lh gs://bucket-name
```

Delete Object

```
gsutil rm gs://bucket-name/file-to-delete
```

Advanced gsutil Operations

Parallel Transfer (for faster bulk operations):

```
gsutil -m cp -r large-local-dir gs://bucket-name
```

Synchronize Directories:

```
gsutil -m rsync -r local-dir gs://bucket-name
```

Copy Between Buckets:

```
gsutil -m cp -r gs://source-bucket gs://dest-bucket
```

8. SUMMARY AND BEST PRACTICES

Key Learnings

Private Google Access and Cloud NAT: Secure connectivity without public IPs

Cloud Storage: Flexible, scalable object storage with multiple storage classes

Cloud SQL: Fully managed relational databases with automated operations

Cloud Spanner: Global SQL with strong consistency for mission-critical apps

BigTable: NoSQL solution for massive-scale time-series and analytical data

gsutil: Powerful CLI for automated Cloud Storage operations

Best Practices Summary

Network: Use Private Google Access for internal calls; Cloud NAT for outbound traffic

Storage: Enable versioning, implement lifecycle policies, use appropriate storage classes

Database: Enable HA for production; use SSL/Cloud SQL Proxy for secure connections

Operations: Use gsutil -m for parallel transfers; monitor costs and performance

Conclusion

You have successfully learned about GCP's core storage and database services. By mastering these services and following the best practices outlined in this guide, you're well-equipped to build scalable, secure, and efficient applications on Google Cloud Platform. Continue exploring advanced features and configurations to deepen your expertise.